PHY-275 Fall 2026 Pop-quiz #5 Lagrange Intuition

1) **In the first task of the pop-quiz you will be asked to derive Euler-Lagrange equation given some coordinates like $q^{*}, q^{\#}, q^{\$}, and\ q^{\%}$. You will be judged by the right amount of DOF or evoke the field theory form (or not).**

2) **In the 2nd part of the pop quiz, you will be asked to drive the equation of motion of a classical mechanical system. Something that can tilt like a rod, or something that can roll like a ball.**